



Calibrant

Interconnection of Large Loads to the Interstate Transmission System

Docket No. RM26-4-000

Calibrant Energy Holdings, LLC (“Calibrant”) supports the action taken by Secretary Wright to address the interconnection of large loads. The letter appropriately recognizes the benefits of large load flexibility and recommends an expedited interconnection study process for loads that are curtailable or co-located with generation or energy storage. We’re pleased that the Commission has responded quickly by opening this docket and by providing an opportunity for public comment. In these comments, we recommend five specific actions that the Commission should take and then provide direct feedback on several of the principles offered in the Advanced Notice of Proposed Rulemaking (“ANOPR”).

In the final rule, the Commission should:

1. Ensure a transparent interconnection process for large loads, where the transmission owners and regional system operators (“RTO/ISOs”) provide customers with the specific constraints, system conditions, and/or the hours that are driving the need for new system upgrades and investment. The clear identification of system needs will allow customers to develop solutions to address those needs. Without visibility into what is causing the system to bind, customers will be unable to develop alternative solutions, leading to delayed

interconnections, economic loss, and potentially an overbuilt, unnecessarily expensive electric system.

2. Require that the transmission owners and RTO/ISOs study load flexibility (including hybrid configurations of load including generation or energy storage) as part of the load interconnection process and identify the amount of flexibility needed to avoid each system constraint identified in the study.
3. Provide a pathway for large load customers to propose, develop, and implement load flexibility solutions (including hybrid configurations of load including generation or energy storage) to defer or avoid the need for, some or all, new infrastructure if the customer commits to providing flexibility during the hours or system conditions identified during the study process. Load flexibility solutions should be considered on equal footing to traditional infrastructure solutions in system planning processes. The Commission should require system planners to include this flexibility into their system planning processes and develop operational tools to dispatch flexibility to match modeled performance.
 - Customers' commitments should be binding for the needs identified by the interconnection study; however, these commitments should not prohibit large load customers or hybrid facilities from providing additional market services (and being appropriately compensated) during other periods when capable of doing so.
4. Recognize that load flexibility (including hybrid configurations of load including generation or energy storage) may also allow a large load to interconnect provisionally while needed infrastructure is built for firm service to create a path for large loads to come online quickly but reliably in the interim. The Commission should require a specific step

at the end of the interconnection study process to identify the capability to connect a large load that is willing to commit to flexibility in a provisional manner, as infrastructure is built to serve large load customers on a fully firm basis.

5. Maintain clear, fair RTO-level cost allocation mechanisms that reflect customers' contributions to the drivers of new infrastructure. While states will maintain jurisdiction over retail rates, RTOs can allocate costs to Load Serving Entities in a manner that better aligns cost causation and cost allocation. The grid will be more affordable if price signals encourage behavior that reduces the need for additional infrastructure.

The wholesale markets have offered a place for customers to bring creative solutions to meet the system's needs cost-effectively. While this marketplace for innovation is available for many market services, a similar opportunity for transmission service does not exist for load customers because of the inflexible and opaque way in which load connection studies are conducted. The Commission is right to exert its authority over the load interconnection process and can establish a new process which will encourage market participants to bring forward the most cost-effective solutions. The Commission can do so by ensuring that all parties have the information they need to develop alternatives and ensuring that these alternatives will be considered on equal footing to traditional rate-regulated transmission investment.

Pacific Gas & Electric's ("PG&E") "Flex Connect" Program offers the Commission a useful model for the transparency they should require of Transmission Owners. In constrained areas of PG&E's grid, PG&E identifies the hours and amount that customers need to curtail (or otherwise provide flexibility from on-site energy generation or storage) if they wish to defer upgrades and interconnect immediately. Figure 1 includes a graphic from a PG&E filing that summarized the program. In this example, the large load must modify its grid usage (to some

degree) from 4PM-11PM in July, August, and September to defer or avoid the need for new infrastructure. By providing this transparency, customers can connect faster to the grid, and the grid becomes more affordable¹. Similar solutions have been developed on a larger scale as well, including Calibrant and Aligned Data Centers’s project in the Pacific Northwest², where a 31 MW / 62 MWh battery energy storage system will enable the facility to come online and scale operations years earlier than would be possible with traditional utility upgrades.

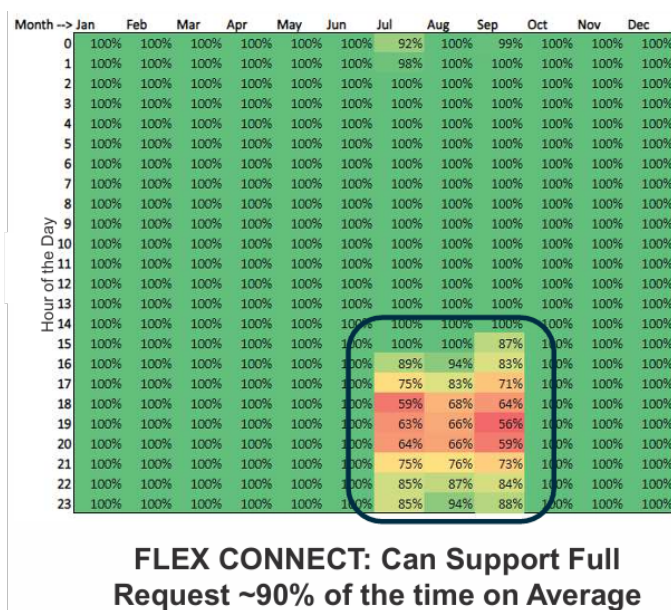


Figure 1: Heatmap from a 2024 filing by the Joint IOUs in California³.

We expect that many other parties will recognize the significance of this proceeding and the opportunity the Commission has to support the next wave of market development. Some have already started. This includes the questions and answers posed by the Nicholas Institute, Former Commissioner Clements, and Roselle, LLP in “*How DOE’s Proposed Large Load Interconnection*

¹ <https://www.pge.com/assets/pge/docs/clean-energy/electric-vehicles/flexible-service-connection-pilot-overview.pdf>

² <https://aligneddc.com/blog/aligned-and-calibrant-to-deploy-first-of-its-kind-energy-solution/>

³ Vehicle-Grid Integration Forum Workshop Report. Filed By Southern California Edison Company (U 338-E), San Diego Gas & Electric Company (U 902 E), And Pacific Gas And Electric Company (U 39 E). Rulemaking 23-12-008. May 21, 2024. P. 64 <https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M532/K262/532262533.PDF>

Process Could Unlock the Benefits of Load Flexibility”⁴, which recognize that the seventh principle in the ANOPR “opened the door to a powerful concept” and that “rules that enable large loads to curtail their consumption of energy from the grid at key times can unlock important benefits, from faster interconnection times to avoided infrastructure and, therefore, lower rates.” The recommendations in this report mirror ours above including creating a clear set of rules to determine what system conditions would require a large load to curtail to avoid or defer transmission upgrades, a study process which incorporates this flexibility, and the ability to obtain non-firm interconnection rights. This report also recognizes that there are multiple meaningful ways to achieve flexibility commitments from customers, including across the “locational stack” as Luminary Strategies calls it in their post on “*DOE’s To-Do List on Load Flexibility Support for Speed to Power*”⁵. Luminary appropriately recognizes that while transmission buildout may remain necessary, in parallel we should be focused on enabling “programs that privilege grid-positive solutions—configurations that make fuller use of the existing grid while the next tranche of wires is being developed,” and clearly outlines that these solutions can be developed in increasingly expansive power footprints from the node the load is interconnecting to a local pocket subject to a common constraint to wider network solutions to the whole-system. The Commission should ensure that the outcome of this rulemaking proceeding creates a viable pathway for grid planners to capitalize on load flexibility from a technical and procedural standpoint, reframing large load additions not as a challenge, but part of the solution to a more efficient grid. Calibrant supports the comments of others in this proceeding, including the Advanced Energy Management Alliance and EmeraldAI, and looks forward to continuing to engage with the Commission in this

⁴ <https://nicholasinstitute.duke.edu/sites/default/files/publications/how-does-proposed-large-load-interconnection-process-could-unlock-benefits-load-flexibility.pdf>

⁵ https://luminarystrategies.substack.com/p/does-to-do-list-on-load-flexibility?utm_source=activity_item



proceeding to develop a process which will better utilize our existing infrastructure and allow the United States to continue to be a leader in data center technology.

Respectfully submitted,

/s/ **Matthew Barnes**

Chief Commercial Officer

Calibrant Energy

Matthew.Barnes@calibrantenergy.com

Calibrant's initial comments on a selection of the 14 principles included in the ANOPR

Principle	Calibrant Comments
1 First, to avoid even arguably affecting the States' jurisdiction over generation facilities, facilities used in local distribution or only for the transmission of electric energy in intrastate commerce, or transmissions consumed by the transmitter. The Commission's jurisdiction should be limited to interconnections directly to transmission facilities, consistent with the Commission's seven-factor test.	The Commission should require each transmission owner and/or RTO/ISO to publish and maintain a list of transmission facilities which meet the criteria of being a transmission facility for purposes of determining jurisdiction of load interconnection, so that market participants know whether their interconnection request would be subject to the Commission's jurisdiction before submitting an application.
3 Third, to the extent practicable, load and hybrid facilities should be studied together with generating facilities. Such an approach will allow for efficient siting of loads and generating facilities and thereby minimize the need for costly network upgrades. For example, siting a large load near or at the same point of interconnection as a new generating facility could reduce the network upgrades needed to interconnect only the load or only the generating facility.	The Commission should avoid intertwining the generation and load interconnection study processes in a way that materially slows the process of interconnecting loads or generation. The Commission could consider expanding the definition of a hybrid facility to load and generation (or energy storage) interconnecting at the same electrical location but not behind a common interconnection point if the load and generation (or energy storage) submit a single interconnection request. Coordinating the interconnection of load and generation (or energy storage) behind a common constraint will result in a more efficient use of the system.
5 Fifth, hybrid facilities should be studied based on the amount of injection and/or withdrawal rights requested. For example, a hybrid facility consisting of a 500 MW load and a 600MW generating facility may seek no withdrawal rights and 100MW of injection rights. This provides incentives for co-location with new generation facilities and ensures efficient buildout of the transmission system.	The Commission should ensure that the study process reflects the expected operating characteristics of the generation (or energy storage) and the load. The study process should provide a pathway for load and hybrid facilities to request that flexibility and specific operating characteristics be incorporated into study assumptions. Solutions that incorporate load flexibility (including hybrid configurations of load including generation or energy storage) should be treated on

Principle		Calibrant Comments
		equal footing as traditional transmission infrastructure solutions. Interconnection rights need to neither be firm nor static for the entire year nor the same every year. Load customers should be able to specify a willingness to be non-firm while infrastructure is being built to accommodate the fully firm project.
7	Seventh, the interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited. The system operator's ability to control such facilities through curtailment and/or dispatch must be sufficient for the system operator to integrate the facility into both operations and system planning. This ensures the timely and orderly addition of large loads to the transmission system in a safe, reliable, and non-discriminatory manner. We seek comment on whether this should be accomplished through a serial interconnection study process or by some other means. We also seek comment on appropriate deadlines for such an expedited study process, including whether such studies can be completed in 60 days.	In addition to expediting the study, the Commission should direct the ISOs/RTOs to expedite the interconnection process for large loads that agree to be flexible. Customers should be allowed to use load flexibility (including hybrid configurations of load including generation or energy storage to enable flexibility) to interconnect provisionally to the system while needed infrastructure is being built to achieve a higher level of transmission service. In order for load flexibility to be accounted for in system planning and operations, customers need to be fully visible to and dispatchable by system operators. However, system operators do not need direct control over customer facilities in order to be confident in their performance for purposes of system planning, similar to other market resources. Large loads should be held accountable for providing flexibility during the time periods where they have agreed to be flexible but should also be allowed to provide other market services to the extent that they are eligible and able to do so. The Commission should study loads and hybrid facilities which are not seeking injection rights serially.
8	Eighth, load and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies. We seek	The Commission should ensure that cost allocation to large load customers for transmission service associated with infrastructure not assigned through the

Principle		Calibrant Comments
	comment on whether such costs should be offset through a crediting mechanism and, if so, over how many years.	interconnection study process is being allocated to those large load customers in a manner that appropriately reflects cost-causation principles. If new large loads are taking transmission service through a FERC jurisdictional interconnection process, then this process should cover all aspects of transmission service.
11	Eleventh, utilities serving large loads, including those at hybrid facilities, should be responsible for transmission service based on their withdrawal rights, as that value amount reflects the quantity of capacity and energy that is being transmitted across the transmission system to the load.	Transmission service should be assessed based on cost-causation principles and not based on maximum withdrawal rights. Transmission service is typically assessed upon customers' contribution to coincident peak demand. The transmission owners and RTO/ISOs should determine which periods or system conditions drive the need for system investment and assess costs based on those periods of time. This creates clear pricing signals to incentivize behavior which will reduce the need for additional investment.
12	Twelfth, utilities serving large loads, including those at hybrid facilities, should be responsible for ancillary services based on peak demand, without consideration of any co-located generation. Any co-located generating facilities will similarly be fully compensated for the provision of ancillary services.	The Commission should be clear on the definition of peak demand. Similar to transmission service, utilities serving large loads should be responsible for services based on cost-causation principles.
13	Thirteenth, there must be a plan to implement these proposed reforms. We seek comment on appropriate transition plans, including the treatment of large load interconnections that are already being studied for interconnection.	The implementation of a new process should not materially slow or hinder the interconnection of large loads that have an interconnection request with their local utility at the time that an RTO/ISO receives approval on compliance associated with any final rule.

CERTIFICATE OF SERVICE

In accordance with 18 C.F.R. § 385.2010, I hereby certify that I have, on this 21st day of November 2025, caused a copy of the foregoing Initial Comments of Calibrant Energy Holdings, LLC to be served on each person designated on the official service list compiled by the Secretary of FERC in Docket No. RM26-4-000.

/s/ Matthew Barnes

Chief Commercial Officer
Calibrant Energy
Matthew.Barnes@calibrantenergy.com